



Can livestock coexist with *Polylepis australis* forests in mountains of central Argentina? Setting thresholds for a land sharing landscape

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ABSTRACT

Developing a land sharing strategy requires a thorough understanding of what is the maximum productivity that can be achieved without altering key ecosystem processes. In the central Argentina highlands, *Polylepis australis* dominates the few remaining patches of forests and its evergreen trees are highly selected by domestic livestock. Grazing exclusion and low herbivory pressure allow *P. australis* forest expansion and soil conservation, while high herbivory pressure promotes forest retraction and environment degradation. However, the extent to which stocking densities should be reduced to allow forest expansion and stand maturation is still unknown. We aimed at answering this question by studying 15 stands located in paddocks with different management, from grazing enclosures to heavily stocked. In 2003 we selected 12 trees less than 2 m tall at each stand. Almost annually along 10 years, for each tree we registered survival, and if alive, we measured the height and two perpendicular diameters of the canopy. We built indicators of vertical and horizontal biomass accumulation rate along the study period and calculated an average effective stocking density (i.e., animal units per ha of non-rocky surface, AU ha⁻¹) for each stand. We found a piece-wise relationship with two pieces between the accumulation rate of each biomass component and average stocking density. Biomass accumulation was maximum under grazing exclusion (i.e. zero stocking density). Between zero and 0.21 AU ha⁻¹ we detected a strongly negative relationship among stocking density and biomass accumulation. Effective stocking densities lower than 0.12 and 0.06 AU ha⁻¹ promoted a widespread vertical and horizontal biomass accumulation, respectively. Stands with stocking densities higher than 0.21 AU ha⁻¹, had null horizontal and vertical biomass accumulation and higher tree mortality. We suggest that effective stocking densities lower than 0.12 AU ha⁻¹ allow a land sharing strategy in central Argentina highlands, ensuring the conservation of a traditional economic activity and the restoration of *Polylepis australis* forests. Additionally, in areas where *Polylepis australis* is absent we suggest active restoration without herbivory pressure. Finally, we suggest that these management recommendations may be studied and promoted throughout the extension of *Polylepis* spp. forests in South America.

1. Introduction

Livestock may prevent woody species regeneration and progressively convert forested lands into grasslands or shrublands; while in treeless landscapes potentially covered by forest, livestock can prevent forest expansion and stand maturation (Cairns and Moen, 2004; Speed et al., 2011; Archer et al., 2017; Bernes et al., 2018; Mazzini et al., 2018). Consequently, it has traditionally been suggested that in many systems, forest conservation and livestock production are incompatible activities. However, during the last years, an increasing number of publications have argued that certain types of agricultural land use and

traditional practices may be compatible with the conservation of forests or forest fragments, thus promoting a land sharing strategy (Phalan et al., 2011; Broom et al., 2013; Fischer et al., 2014).

Even though plants can experience a negative impact from large domestic herbivores, some species can either avoid (e.g. through physic and chemical defense) or compensate (e.g. through fast regrowth) tissue removal. However, plant response is conditioned to a large extent by the evolutionary context (Belsky, 1986; Bernes et al., 2018; Mazzini et al., 2018; Öllerer et al., 2019). Thus, in a given ecosystem, extrapolation of management rules developed for other systems may be misleading; while management plans based on short-term local studies

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might be biased because of the particular climatic conditions of the study period (Noy-Meir, 1993; Archer et al., 2017), a point that was especially highlighted in recent reviews of studies around the world (Bernes et al., 2018; Mazzini et al., 2018; Öllerer et al., 2019). Therefore, the promotion of a land sharing strategy including forest conservation and cattle production requires long-term studies to obtain a sound understanding of tree species response to different herbivory pressures.

Livestock production is one of the most important economic activities in mountain ecosystems (Kohler et al., 2006). However, domestic herbivores may produce significant impacts on vegetation (Cairns and Moen, 2004; Speed et al., 2011), even in ecosystems which evolved with large herbivores as is the case of South America, where grazing pressure in extensive agricultural systems is one order of magnitude greater than in wildlife-dominated ecosystems (Oesterheld et al., 1992). For example, in the southern extreme of Brazil, the eastern edge of the highland plateau is covered by a mosaic of *Araucaria angustifolia* forests and Campos grasslands in which the expansion of *A. angustifolia* into the adjacent grassland is prevented by domestic grazing pressure and recurrent fires (Behling and Pillar, 2006).

Polylepis forests are endemic of South American highlands. These forests span across South American mountains from the tropical and subtropical Andes to Central Argentina. In spite of their extensive latitudinal cover, the huge anthropogenic pressures they are experiencing (Cuyckens and Renison, 2018; Renison et al., 2018), and their fundamental role in biodiversity conservation and the provision of a number of ecosystem services (Lloyd and Marsden, 2008; Cranford and Mourato, 2011; Poca et al., 2018a,b), *Polylepis* forests are surprisingly understudied in comparison with other South American forests, such as those dominated by *Nothofagus* spp (Renison et al., 2018). At the southernmost end of *Polylepis* distribution, part of the highlands of central Argentina above 1700 m a. s. l. are dominated by patches of *Polylepis australis* Bitter, immersed in a grassland landscape with granitic outcrops. These evergreen forest fragments are generally restricted to sites far from sources of human disturbances and to steep slopes in mid to low topographic positions, where fire and livestock have relatively low incidence (Cingolani et al., 2004, 2008; Renison et al., 2006, 2015). Even though in these highlands there are no physiological restrictions to the establishment of forests at any topographic position (Cingolani et al., 2008; Pais Bosch et al., 2012; Marcora et al., 2013; Renison et al., 2015), forests and shrublands dominated by *P. australis* occupy only 12% of the landscape, while another 30% of the area has very sparse and generally small individuals (Cingolani et al., 2004).

Polylepis australis forests in central Argentina harbour a unique flora, historically related to the south tropical Andes and with a high number of endemic species (Cabido et al., 2018). These forests are located in the headwater catchments that provide water to over three million people in the lowlands (Cingolani et al., 2015; Poca et al., 2018a,b) and therefore, they have high conservation value. However, only the 11% of the land above 1700 m asl, which is the habitat of *Polylepis australis* forests, is under exclusive government management protected by a National Park, while the rest is under private decision. At the same time, about a half of the private lands belong to a provincial reserve, but this protection status is currently weak and most of the lands are subjected to forest degradation and soil erosion (Cingolani et al., 2004, 2008, 2013). In 2017 a new Provincial Law (N° 10467) was set to compel private owners to allocate 10% of their lands to be covered with forest in a time lapse of 10 years. Consequently, in this new scenery adequate management of herbivory pressure is essential to protect and restore forest cover.

During the last decade, scientific evidence in the area suggests that low herbivory pressure or grazing exclusion allows *P. australis* forest expansion and stand maturation, while high herbivory pressure promotes forest retraction (Teich et al., 2005; Giorgis et al., 2010; Cingolani et al., 2014). However, the extent to which stocking density

should be reduced to allow forest expansion is still unknown. Furthermore, there is an urgent need to change the current grazing management of the area, since it is associated with the degradation of the natural and social capital (Renison et al., 2010; Cingolani et al., 2013; Poca et al., 2018a,b). In this context, we aimed at contributing to a land sharing strategy in this conservation-priority area which is under livestock production. For achieving this, we conducted a long-term 10-year study to estimate the maximum stocking density which allows a widespread *Polylepis australis* forest expansion and stand maturation at central Argentina highlands. We also expect to set basis for management policies useful for other *Polylepis* forests across the Andes, which are under similar anthropogenic pressures.

2. Material and methods

2.1. Study area

The study was conducted in the upper altitudinal belt (1700–2800 m a.s.l.) of the Córdoba mountains in central Argentina (31°34' S, 64°50' W). At 2100 m a.s.l., mean temperatures of the coldest and warmest months are 5.0 and 11.4 °C, respectively, with no frost-free period. Mean annual precipitation is 921 mm (1992–2017), with most rainfall concentrated in the warmest months, between October and April (Colladon and Pazos 2014; Colladon, 2018). The main economic activity is livestock rearing (mainly cattle, but also sheep, horses and goats), which began early in the seventeenth century. *Lama guanicoe* was the characteristic native herbivore in the area, and one of the most important sources of food for humans until 6000 – 3000 AP. *L. guanicoe* was accompanied by *Hippocamelus* sp., *Ozotoceros bezoarticus* and probably *Rhea americana* and *Pterocnemia pennata*. However, they all became locally extinct at the end of the 19th century (Medina and Rivero, 2007).

The Quebrada del Condorito National Park is located in the upper portion of the Córdoba mountains (31°34'S, 64°50'W) in central Argentina, covering an area of 24,774 ha. Since the implementation of the Park in 1996, livestock pressure was maintained at low to moderate levels in some areas, while it has been completely excluded in others. Additionally, in 2007 a reintroduction project of *Lama guanicoe* was implemented, in order to recover the population of the native herbivores but with small success until present (Barri 2016). This was done to prevent plant diversity loss and the excessive plant biomass accumulation which is a potential source of fuel for wildfires (APN 2007). Above 1500 m a. s. l. and outside the national park, part of the area belongs to the Pampa de Achala provincial reserve (Fig. 1), whose lands are under private management and, in general, subjected to high herbivory pressure and fire frequency (Cingolani et al., 2008, 2013, 2014).

Vegetation in this area is a mosaic of tussock grasslands, grazing lawns, fernlands, granite outcrops, evergreen *P. australis* forests and eroded areas with exposed rock surfaces. In this mosaic, stands dominated by *P. australis* trees or shrubs represents only the 3% of the area, while another 9% consists in open forests or shrublands with 5–55% *Polylepis* cover. An additional 30% of the area, dominated by tussock grasslands and granite outcrops, harbors very sparse *P. australis* individuals (< 5% cover) (Cingolani et al., 2004; Poca et al., 2018a). Stands within open forests/shrublands or in areas with sparse *P. australis* individuals are characterized by small trees with heights generally below 116 cm. In turn, within the 3% of the area dominated by *P. australis*, well conserved stands with individuals higher than 7 m are extremely rare, being 14 m the highest recorded (Renison et al., 2011).

An additional threat to these forests is that during the last decade woody invaders (e.g. *Cotoneaster franchetii*, *Rosa rubiginosa*, *Rubus ulmifolius*) are expanding from the lowlands towards the upper belt potentially covered by *P. australis* (Giorgis et al., 2011, 2016, 2017).

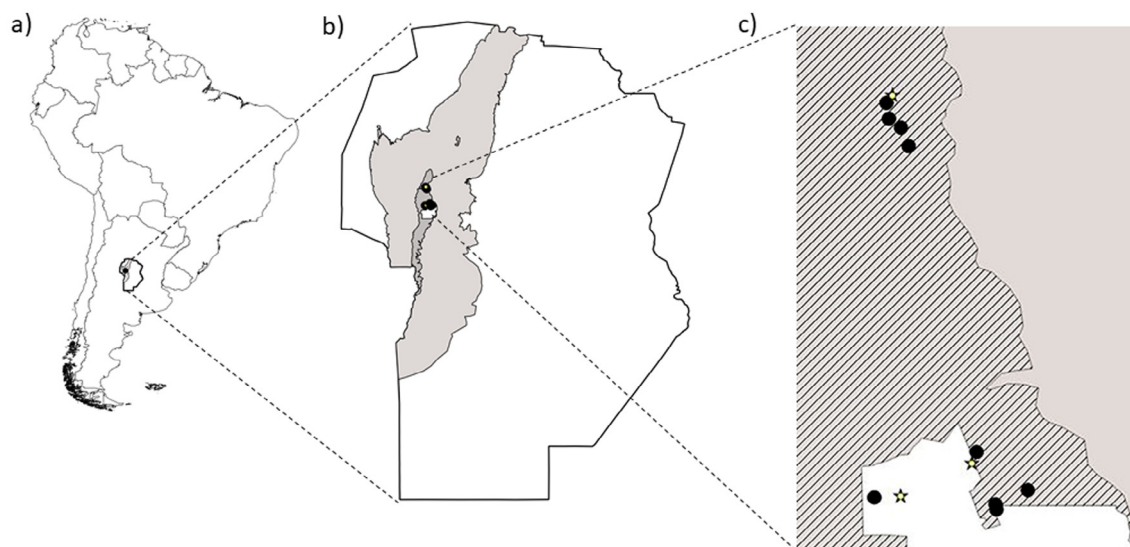


Fig. 1. Location of (a) Córdoba province in Argentina, South-America, (b), Córdoba mountains, light grey the area between 500 and 1700 m a.s.l. and dark grey the area above 1700 m a.s.l. (c) stands located in the study area with variable stocking density indicated with black circles and the three enclosures built in 2004 indicated with stars; the white area represents part of the National Park Quebrada del Condorito, the striped area in black grey is the mountain zone under private ownership above 1700 m a.s.l.

2.2. Sampling design and field measures

We selected 15 *P. australis* stands (0.25–0.60 ha) with low tree density ($< 10\%$ cover), with an average of 643 trees ha^{-1} , mostly consisting in small individuals (< 2 m tall) (Teich et al., 2005), distributed in the central and northern portion of Córdoba mountains. Three stands were enclosures without grazing. These were constructed in January 2004, two inside the National Park and one outside of it, all three of them surrounded by areas under livestock pressure. A fourth stand was inside the National Park in an area without livestock pressure since 1999, while the remaining 11 stands were under livestock pressure with different stocking densities, both inside and outside the Park (Teich et al., 2005; Giorgis et al., 2010). In each stand, we selected 12 trees less than 2 m tall within the reach of livestock consumption. At one stand, the tag of one tree was lost and it had to be discarded from the study. Thus, the study comprised a total of 179 trees.

Every year from 2003 to 2013 (except in 2004 and 2007), we measured the height and two perpendicular diameters of the canopy for each selected living tree, and if not living, we registered it as dead. For standardization, we made all measurements in August, before the beginning of the growing season, which goes from October to April (Giorgis et al., 2010). Additionally, at each sampling date and stand we measured the frequency of cattle, horse, sheep and goat dung depositions, by randomly placing 100 times a 30 cm $\times$ 30 cm square (von Müller et al., 2017).

2.3. Data processing and analyses

2.3.1. Biomass components

We estimated the vertical and the horizontal biomass components for each stand and year. To estimate the vertical component, we summed up the height of all selected living trees per stand (cm). To estimate the horizontal component, we used the two perpendicular canopy diameters to calculate the cover (cm^2) per selected living tree, assuming an elliptical shape and then summed up the cover of all living trees in each stand and year. To standardize data, for each year we divided both sums (height and cover) by the initial number of trees in each stand. As we worked with a fixed number of selected trees, our estimations did not consider biomass additions due to the establishment of new individuals. However, the establishment of new individuals is

generally small and represents a reduced value in terms of biomass at the temporal scale of this study (Renison et al., 2015).

2.3.2. Biomass accumulation rate and tree survival

For each of the 15 stands we calculated the accumulation rate of the vertical and horizontal biomass components (cm year^{-1} and $\text{cm}^2 \text{year}^{-1}$, respectively) as the slope of a linear regression between time and biomass for each component. Positive values of slopes indicate an increase in biomass during the period, negative values indicate a reduction in biomass, and close to zero values indicate that biomass remained relatively constant. Additionally, we calculated the survival rate per stand as the proportion of living trees at the last measurement date.

2.3.3. Stocking density

For each stand, we calculated an average stocking density in the 10-year period under study. First, for each stand and year, we converted cattle dung frequency data (dung frequency of other animals was minimum) into a value of stocking density using the model fitted and validated for our study area by von Müller et al. (2012) as follows:

Stocking density = $0.017 \times \text{dung frequency}$

In which stocking density is expressed as Animal Units per total area and dung frequency is expressed as the proportion (%) of the 100 squares which had cattle dung. Then, for each stand and year we calculated the effective stocking density, which is the animal density per ha of non-rocky surface (hereafter AU ha^{-1}). The rock cover for each stand was obtained from data previously reported by Teich et al. (2005). Finally, for each stand we averaged the stocking density of all years and obtained the average effective stocking density during the whole study period (Table 1).

2.3.4. Biomass vs stocking density

To estimate a stocking density which guarantees a widespread positive biomass accumulation rate, we firstly explored the relationship between the accumulation rate of each biomass component and stocking density through graphic tools (scatterplots). For both biomass components we observed a piece-wise relationship with two pieces, which is a model where the relationship between the response and the explanatory variable is represented by two straight lines connected at

Table 1

Annual vertical and horizontal biomass accumulation rate per initial tree in each of the 15 stands through the ten-years period under study. Surviving trees at the last date and the average effective stocking density per stand are shown with their minimum and maximum annual values in parentheses.

Stand ¹	Vertical accumulation (cm year ⁻¹ tree ⁻¹)	Horizontal accumulation (cm ² year ⁻¹ tree ⁻¹)	Surviving trees (% survival)	Effective stocking density (AU ha ⁻¹) ²
13	12.22	661.8	12 (100)	0
14	6.39	228.3	12 (100)	0
15	15.75	320.5	11 (100)	0
9	8.63	33.66	12 (100)	0.02 (0–0.06)
1	9.16	37.61	11 (91.6)	0.11 (0.05–0.22)
12	5.55	6.37	9 (75)	0.11 (0.05–0.16)
11	6.237	20.78	12 (100)	0.13 (0.1–0.4)
6	1.46	5.44	12 (100)	0.22 (0.09–0.35)
2	–2.34	–0.67	10 (83.3)	0.26 (0.12–0.29)
5	0.08	0.620	11 (91.6)	0.27 (0–0.16)
8	–1.77	–0.50	8 (66.7)	0.31 (0.24–0.37)
10	–0.08	3.30	10 (83.3)	0.45 (0.45–0.61)
3	–0.11	1.21	8 (66.7)	0.55 (0.57–0.93)
7	–1.73	0.22	8 (66.7)	0.55 (0.12–0.91)
4	1.82	5.18	12 (100)	0.67 (0.32–1.1)

¹ We used 12 trees per stand were used to calculate the variables in this table, except for stand 15 which had 11 studied trees. Stand 13, 14 and 15 are enclosures.

² Effective stocking density, the rock proportion was discounted from the total area in each stand.

breakpoint value (Muggeo, 2008). The first piece (i.e. stands with low stocking densities) appeared as a negative linear relationship, and the second piece (i.e. stands with higher stocking densities) as a constant relationship. Therefore, for each biomass component we fitted a linear regression for the first piece with the accumulation rate as the dependent variable and the stocking density as the independent one. To select the best set of data to estimate that linear function, we ranked the stands from the lowest to the highest stocking density and performed a first linear regression including the six stands with lowest stocking density. Then we added one additional stand at a time and perform a new linear regression. We continued to include stands until maximizing the r^2 -value. From the selected linear regression, we estimated the linear function and its 95% confidence intervals for the stands and for the mean. From the confidence intervals, we obtained the highest stocking density values whose lowest limit, either for the individual stand and for the mean, were above zero.

For each component, we also calculated the average accumulation rate and the confidence intervals for the constant part of the relationship (i.e. the second piece). This was done by averaging the stands with highest stocking densities and calculating their confidence intervals. To select the points to use for this average, we first calculated the average for the last three points. Then we began to add backward one point at a time, until minimizing both the confidence interval for the mean and for the individual stand.

We analyzed if survival varied with stocking density through a generalized linear model using a binary distribution and a logit link function with each tree as a case, the stand as a random factor and stocking density as a fixed factor.

We were aware that herbivory pressure previous to our experiment, which was generally similar to the stocking rate during the experiment, may have conditioned the accessibility and initial size of our selected trees. To account for this, we explored the potential bias in tree selection at the beginning of our study to aid in the interpretation of results. For this, we analyzed through Pearson correlations if we tended to select smaller (i.e. less biomass in both components) and more inaccessible trees in stands under high stocking density than in stands under low stocking density. Accessibility at the stand level was calculated as the average of the accessibility at the tree level; each tree coded from 1 (low accessibility) to 4 (high accessibility) taken from Teich et al. (2005). We excluded the three stands within the enclosures built in 2004.

All data analyses were performed using the R software package, version 3.4.2 (R Development Core Team, 2017) and Infostat (Di Rienzo et al., 2016).

3. Results

The annual biomass accumulation rate of the vertical component (standardized per initial tree) ranged from -2.34 to 15.75 cm yr⁻¹ tree⁻¹, whilst the horizontal component rate varied between -0.67 and 661.8 cm² yr⁻¹ tree⁻¹. Dead trees varied between zero and four (33.3%) per stand. Stocking density was zero for the three enclosures, as expected, and almost zero for the stand in the National Park without livestock (0.02, this small, non-zero value is because occasionally some cattle trespassed the fences from the neighbour private establishment). For the remaining stands, the stocking density varied from 0.11 to 0.67 AU ha⁻¹ with small variations in each stand throughout the study period (Table 1).

3.1. Biomass components vs stocking density

For the vertical component of biomass, the linear regression with the highest regression coefficient ($r^2 = 0.82$) included 11 stands ranging from zero to 0.31 AU ha⁻¹ (Fig. 2), in which the stocking density had a significantly negative effect ($p < 0.0001$). At 0.12 AU ha⁻¹ the lowest limit of the individual stand confidence interval was zero, while at 0.21 AU ha⁻¹ the lowest limit of the confidence interval for the mean was zero (Fig. 2). The mean vertical biomass accumulation rate for the eight stands with stocking densities higher than 0.20 AU ha⁻¹ (i.e. those selected to evaluate the constant part of the relationship), was -0.33 cm year⁻¹ tree⁻¹ and the confidence interval for the mean ranged from -1.61 to 0.94 , while for the individual sample it ranged from -4.7 and 4.0 cm year⁻¹ tree⁻¹.

The same eleven stands were selected for the linear regression of the horizontal biomass component, having the stocking density a statistically significant negative effect ($p = 0.0028$, $r^2 = 0.65$, Fig. 3). At 0.06 AU ha⁻¹ the lowest limit of the individual stand confidence interval was zero, while at 0.21 AU ha⁻¹ the lowest limit of the confidence interval for the mean was zero (Fig. 3). The mean horizontal accumulation rate for the eight stands with stocking densities higher than 0.20 AU ha⁻¹ (i.e. those selected to evaluate the constant part of the relationship), was 1.85 cm² year⁻¹ tree⁻¹ and the 95% confidence interval for the mean ranged from -0.21 to 3.91 cm² year⁻¹ tree⁻¹, while for the individual sample it ranged from -5.2 to 8.9 cm² year⁻¹ tree⁻¹.

3.2. Trees survival vs stocking density

Tree survival per stand was negative and significantly affected by stocking density ($F = 4.33$, $p = 0.04$). Survival was 100% for the four stands with zero or almost zero stocking densities and decreased up to

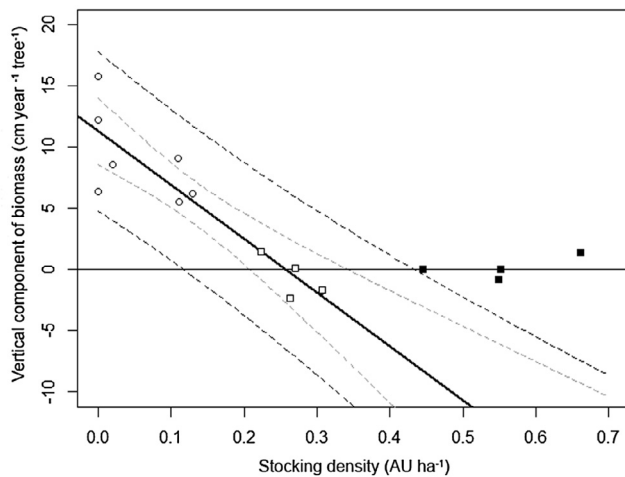


Fig. 2. Vertical component of biomass ($\text{cm year}^{-1} \text{ tree}^{-1}$) as a function of effective stocking density (AU ha^{-1}). Open symbols (circles and squares) represent the selected stands with the highest R^2 (0.82) for the first part of the piece-wise relationship (linear regression: $y = -44.02x + 11.31$), solid black squares represent the discarded stands for the regression analysis. The solid black line represents the regression line. The grey dashed line represents the 95% confidence interval for the mean, while the black dashed line corresponds to the 95% confidence interval for the first part of the piece-wise relationship. Squares (both open and black) were used for estimating the accumulation rate average for the constant part of the piece-wise relationship. The constant line (second piece) was not plotted to improve visual understanding.

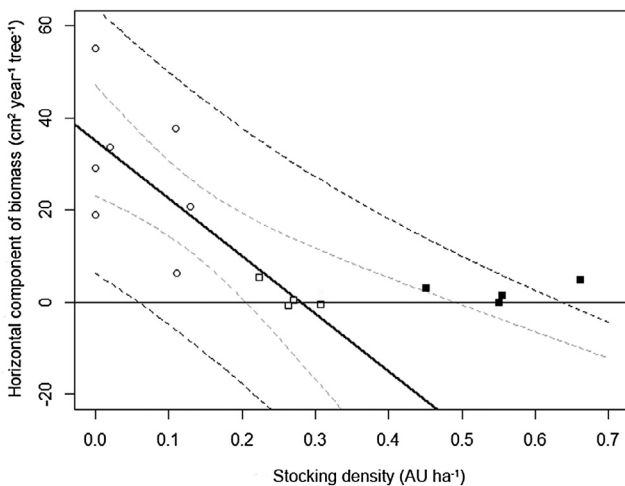


Fig. 3. Horizontal component of biomass ($\text{cm}^2 \text{ year}^{-1} \text{ tree}^{-1}$) as a function of effective stocking density (AU ha^{-1}). Open symbols (circles and squares) represent the selected stands with the highest R^2 (0.65) for the first part of the piece-wise relationship ($y = -125.36x + 35.112$), solid black squares represent the discarded stands for the regression analysis. The solid black line represents the regression line. The grey dashed line represents the 95% confidence intervals for the mean, while black dashed line corresponds to the 95% confident intervals for the first part of the piece-wise relationship. Squares (both open and black) were used for estimating the accumulation rate average for the constant part of the piece-wise relationship. The constant line (second piece) was not plotted to improve visual understanding.

67% with high stocking densities (Table 1).

3.3. Stocking density vs stand tree characteristics

For the 12 stands which were not recent enclosures, we found a negative significant relationship between the initial vertical biomass component and the stocking density ($r = -0.67$, $p = 0.02$, $n = 12$),

and a non-significant negative trend between the initial horizontal biomass component and stocking density ($r = -0.45$, $p = 0.14$, $n = 12$). Additionally, we found a marginally significant negative relationship between stocking density and average tree accessibility for herbivores ($r = -0.54$, $p = 0.07$, $n = 12$). All results indicate that stands with high stocking densities during the study period tended to have less biomass and less accessible trees at the beginning of the study than stands with lower stocking densities.

4. Discussion

Polylepis forests are distributed through the Andes and contiguous mountains along South America. These forests distribution is constrained to patches partly due to anthropogenic factors (Kessler, 2002). In central Argentina highlands, after more than two centuries of live-stock and fire mismanagement, *Polylepis australis* forests occupy a low proportion of the landscape. Through this study we aimed at understanding the extent to which stocking densities should be reduced to allow forest expansion and stand maturation. Our main results show that in *Polylepis australis* stands with low tree density (< 650 trees ha^{-1}), a stocking density lower than 0.12 AU ha^{-1} will promote a positive vertical biomass accumulation, while a stocking density lower than 0.06 AU ha^{-1} will promote both a vertical and horizontal expansion, favouring the maturation of forests. These results are in line with previously reported ones by Cingolani et al. (2014) in a study at the landscape scale. These authors suggested an expansion in forest cover at paddocks with stocking densities below 0.12 ha^{-1} . Our results also support preliminary findings which suggested that *P. australis* might compensate grazing pressure at low stocking densities (Giorgis et al., 2010). Thus, small *Polylepis australis* trees can grow in height while coexisting with livestock production at low stocking densities ($< 0.12 \text{ AU ha}^{-1}$), but to guarantee the horizontal expansion they need even lower stocking densities at the stand level. These stocking density values are far below the traditional grazing management in the area, that usually goes from 0.22 to 0.34 AU ha^{-1} ; even in some cases stocking densities may attain more than 1 AU ha^{-1} (Teich et al., 2005). These grazing pressures are normally associated with soil erosion and forest degradation (Cingolani et al., 2013, 2014), suggesting that traditional cattle management and forest conservation are incompatible activities. Altogether, we can conclude that our system is compatible with native or domestic herbivores with stocking densities lower than 0.12 AU ha^{-1} . Cattle and native domestic and wild large herbivores (e.g. llama and guanaco) are commonly present along the whole *Polylepis* forests distribution. However, their effect on forest establishment and regeneration have only been studied in central Argentina (Renison et al., 2018). This scarcity of studies, likely reflect that herbivory has not been identified as a serious conservation issue, together with a reduced budget on scientific studies in other countries with *Polylepis* forest as Bolivia and Peru (Cimoli et al., 2005). Still, in coincidence with Renison et al. (2018), we especially highlight the necessity of similar studies throughout the Andes, given they could not only improve conservation projects' success, but also the promotion of a land sharing management strategy along the complete distribution of *Polylepis* forests.

We found drastic changes in *Polylepis australis* biomass accumulation rate along the evaluated gradient of stocking density. Between null to moderate (0.21 AU ha^{-1}) stocking densities, we observed a strong negative relationship for both biomass components, while above 0.21 AU ha^{-1} the accumulation in the vertical and horizontal components was null (not significantly different than zero). This pattern indicates that between zero and 0.21 AU ha^{-1} , biomass accumulation is highly sensitive to livestock pressure. That is, a small variation of herbivory pressure produces a high impact on *Polylepis australis* growth. Among both extremes of this range of stocking densities, biomass accumulation rate decreased ten-fold, which represents the higher registered impact on growth for a woody species in central Argentina. For instance, at the

lower and more xeric part of our mountain range, Capó et al. (2016) showed a four-fold decrease in growth between livestock exclusion and a stocking density of 0.21 AU ha^{-1} for two common alien species and a non-significant difference in growth for four native woody species. For the same study area, Torres and Renison (2015) found a comparatively similar small variation in growth of planted saplings of other two native species. Additionally, our results indicate that over 95% of the stands have a positive biomass accumulation when stocking densities are lower than 0.12 or 0.06 AU ha^{-1} for the vertical and horizontal component, respectively. Between $0.12/0.06$ to 0.21 AU ha^{-1} a positive biomass accumulation is still expected when averaged across stands, but at the individual stand level the result is very variable, indicating that some stands may not experience vertical or horizontal growth at all.

In stands with stocking densities higher than 0.21 AU ha^{-1} , biomass accumulation is null and constant for both components. It may seem striking that biomass remains relatively constant with no decline even with very high stocking densities, considering the high selectivity for this species and its importance as winter forage (Giorgis et al., 2010). We suggest that this pattern was caused by the high herbivory pressure of those stands supported during years before the start of the present study. As a consequence of previous high herbivory pressure, survivor trees were small at the beginning of our study, with part of their biomass protected from browsing, as indicated by their low accessibility indices (i.e. negative relationships of stocking density with initial biomass and accessibility, see pictures in Appendix S1). The low accessibility is caused mainly by rocks surrounding the individual. That hidden portion of the tree may produce more biomass which became accessible to animals and is immediately consumed, as was reported by Giorgis et al. (2010). Those trees could survive in that partially inaccessible sites (Teich et al., 2005; Renison et al., 2011, 2015); though, as we report here, they do not accumulate aerial biomass, and in the long term they are subjected to higher tree mortality. Inaccessible sites constitute small geologic refuges (i.e. rock outcrops, cliffs, etc., sensu Milchunas and Noy-meir, 2002) in which plants avoid grazing. Small geologic refuges are common in mountain systems and have been suggested to have two main roles for conservation and management (Milchunas and Noy-meir, 2002). First, refuges constitute an essential source of propagule pressure which may guarantee forest expansion if grazing pressure is reduced in the surrounding landscape, assuring its potential expansion. Second, they are microhabitats important for promoting the diversity of the overall plant community at the landscape (beta diversity) scale and the presence of rare or endangered plant species at local scale (alfa diversity). In the case of our study system, there is still a need of further research to unravel the specific role of these *Polylepis* forest refuges at the landscape scale.

The fact that the stocking density that would ensure a positive horizontal biomass accumulation rate for the 95% of the stands was very low (0.06 AU ha^{-1}), indicates that the horizontal expansion is more variable and threatened than the vertical expansion (i.e. the difference between the highest and the lowest value in the confidence interval is higher in the horizontal component than in the vertical one). Typically, if distal shoots are cut off, the apical control or apical dominance is removed (Wilson 2000) and thus, lateral growth is activated in different directions (e. g. Vila et al., 2003; Hoppe-Speer and Adams, 2015). However, the direction of this lateral growth is conditioned by livestock accessibility (Teich et al., 2005; Giorgis et al., 2010), determining different tree shapes which are intimately related to their microhabitat (see pictures in appendix S1).

The piece-wise relationship between both biomass components and livestock density highlights the importance of understanding livestock impact as a continuous response instead of a binary (grazed vs ungrazed) or categorical relation (low, moderate and high), as was reinforced by recent reviews (Mazzini et al., 2018; Öllerer et al., 2019). This type of response is common in nature (e.g. Betts et al., 2007; Wanget al., 2016; Ollerton et al., 2014) because usually, systems have

thresholds which are detectable only if we study the complete response gradient.

5. Management recommendations

Currently, the landscape in central Argentina highlands is the result of more than two centuries of high livestock pressure and fire incidence (Cingolani et al., 2014). This area with no physiological restriction for forest growth (Marcora et al., 2013; Renison et al., 2015) is dominated by grasslands and by rock patches result of soil erosion with a low cover of generally small *Polylepis australis* trees, usually confined to geologic refuges (Cingolani et al., 2004, 2008; Renison et al., 2011 and see pictures in Appendix S1). During the last decade different studies have shown that traditional management is not sustainable and inevitably results in soil loss (Renison et al., 2010, Cingolani et al., 2013, 2014) and a drastic reduction of fundamental ecosystem services such as water provisioning (Poca et al., 2018a,b). In this context, and considering the new provincial legislation which promotes the increase in forest cover, our results bring light on management practices, allowing to provide the following recommendations:

1. **If forest expansion is the aim**, we suggest livestock exclusion because most *Polylepis australis* trees in the highlands are small (Renison et al., 2011), and without herbivory they may grow as much as $15 \text{ cm year}^{-1} \text{ tree}^{-1}$ in the vertical component and $600 \text{ cm}^2 \text{ year}^{-1} \text{ tree}^{-1}$ in the horizontal component. These results are coincident with those reported by Renison et al. (2015), who showed tree growth rates between 4 and 17 cm per year under grazing exclusion. Thus, in 15–20 years a small tree may reach the browsing limit and escape from cattle pressure. Additionally, active restoration (plantation of seedlings) might also be necessary, since this species has a very low dispersal capability (6–7 m from the paternal tree, Torres et al., 2008) and good reproductive individuals are absent in most of the area (Renison et al., 2011).
2. **If livestock production with forest conservation is the aim**, we suggest stocking effective densities lower than 0.12 AU ha^{-1} for vertical biomass accumulation. Even though this stocking density does not guarantee a horizontal biomass accumulation, trees may eventually escape in height and then extend in canopy cover above the browsing limit over the years. It is important to highlight that our results are based on localized stand-level stocking densities and not on paddock stocking densities. At the stand level, the local stocking density may be higher or lower than the paddocks' one because stand characteristics may attract or repel herbivores (von Muller et al., 2017). In our study area, von Muller et al. (2017) found that cattle actively select lawn patches dominated by short palatable plants and avoid rough patches and steep terrains. In coincidence with these results, Cingolani et al. (2014) found that the advancement of woody cover is facilitated on steep slopes and their retraction is favoured on gentle ones. As management decisions are necessarily taken at the paddock level, their characteristics may need to be considered. Even a paddock with stocking densities below 0.12 AU ha^{-1} might have some small stands with *Polylepis australis* trees under high browsing pressure. Conversely, paddocks with higher stocking densities may have some stands in steep slopes with local low stocking densities where the forest maturation would be possible. Moreover, and in agreement with Cingolani et al. (2014), we emphasize that livestock densities higher than 0.21 AU ha^{-1} are totally incompatible with forest conservation and are clearly associated with soil loss in this system (Cingolani et al., 2013). Consequently, herbivory pressure higher than 0.21 AU ha^{-1} should not be currently allowed in this area.

Finally, due to the current low cover of *Polylepis australis* forests in good conservation status (Cingolani et al., 2004; Renison et al., 2011) and their disproportional importance for water provisioning service for

the surrounding lowlands (Poca et al., 2018a,b), management strategies must be promoted for passive and active forest restoration. At the same time, it is important to maintain different physiognomies at the landscape scale, especially lawn patches, to promote plant β diversity (Cingolani et al., 2014), bird diversity (García et al., 2008) and control the fast advance of woody alien species from the lowland area (Giorgis et al., 2011, 2016, 2017; Capó et al., 2016, Tecco et al., 2016). Interestingly, livestock pressure has been shown to be an additional tool to control alien expansion in the lowlands of central Argentina (Capó et al., 2016); however, this needs to be specifically investigated for our study highlands. Given *Polylepis* forests along South America experience similar vulnerabilities to anthropogenic factors, management strategies discussed here should be studied and applied for other *Polylepis* species forests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2019.117728>.

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